



**FRONTLINES
EDUTECH**
Building Trust & Careers

DEVOPS WITH MULTI-CLOUD (AWS+AZURE)



FRONTLINES EDUTECH PRIVATE LIMITED



ABOUT US

Frontlines Edutech Private Limited envisions bridging the gap between academia and industry. We understand the pain of unemployment and are trying to wipe off the words of impossible and unemployment from the brains of youth by training and turning them into diamonds. We provide tailor-made, hands-on, and need-based programs on par with industry standards. We believe that the youth of today is the future of tomorrow. And so our courses are tutored by experts from top industries who are passionate about mentoring the youth. Since its inception, Frontlines Edutech has earned the trust and gained recognition from thousands of learners across the state.

OUR MISSION AND VISION

At Frontlines Edutech Private Limited, our mission is to bridge the gap between academia and industry by offering practical, customized training programs that equip youth with the skills needed for easy employment. We aim to eliminate unemployment by fostering a mindset of possibility and transforming learners into skilled professionals. With expert industry mentorship, we empower the next generation with the tools to succeed in today's competitive world.

It is in our vision that each young person be prepared with the skill, confidence, and opportunity to succeed in their career. We want to lead an online skills development platform that provides focus on determining the needs of industries through education, so that it would be well-prepared in nurturing professionals who will shape the future of employment. Innovative training eliminates unemployment and helps in unlocking the potential of thousands of learners.



WHY CHOOSE US

We are Success Trainers, Mentors, and Motivators, passionate about helping you reach your full potential with a blend of warmth, wisdom, positivity, and proactivity. Our goal is always your highest good and greatest growth, offering support that is both caring and daring, fun and focused. We ensure learning is not only powerful but also enjoyable, making every step of your growth journey meaningful and engaging.



Industry Standard Training

Solution-Focused

Results-Driven

Affordable

Transparent

Experience

Quality



DEVOPS WITH MULTI CLOUD (AWS+AZURE) SYLLABUS

OS: LINUX

- Linux Intro
- System commands
- Hardware commands
- File commands
- User commands
- Permission commands
- Search commands
- Editors (VIM)
- Networking Commands
- Software commands
- Shell Scripting on Bash

AWS SERVICES

- Elastic Compute Cloud (EC2)
- Virtual Private Cloud (VPC)
- Simple Storage Service (S3)
- Elastic Block Storage (EBS)
- Identity and Access Management (IAM)
- Relational Database Service (RDS)
- Elastic Load Balancer (ELB)
- Auto Scaling Group (ASG)
- Elastic File System (EFS)
- Elastic Container Registry (ECR)
- Elastic Container Service (ECS)
- Elastic Kubernetes Service (EKS)
- Simple Notification Service (SNS)
- Code Build
- Code Deploy
- AWS Code Pipeline
- Cloud Watch
- Route 53

AZURE SERVICES

- Azure Active Directory
- Virtual Machines
- VM Scale Set
- Azure Load Balancers
- Azure Container Registry
- Azure Container Interface
- Azure Kubernetes Service
- Azure Pipelines
- Azure SQL Database
- Azure Blob Storage
- Azure Discs
- Azure Atrifacts
- Azure Key Valut
- Azure CLI

SCM TOOLS - GIT

- What is VCS
- VCS History
- SCCM
- Revision Control System
- Subversion
- Concurrent Versions System
- Why Git
- Git Stages
- Working Directory
- Staging Area
- Repository (Local, Central, Remote)
- Git Installation
- Git Lifecycle
- Git Logs
- Git Push



- Git Pull
- Git Cloning
- Git Branch
- Git Merge
- Git Stash
- Git Tags
- Git Cherry Pick
- Git Revert
- Git Merge
- Merge Conflicts
- Configuration of User
- Ignoring Content
- Branching Strategies

GITHUB

- Git Repos (Private & Public)
- Integrating Repos
- Forking
- GitHub FileAdd
- Tokens
- Compare & Pull Request
- Renaming Repos
- Danger Zone Options
- Making Public Repo as Private
- Ownership Transfer
- Archiving Repos
- Deleting Repos
- Accessing the Private Repos
- Advantages & Disadvantages
- Difference Between Git and Other Tools
- Interview Questions

BITBUCKET

- Bit Bucket Repos (Private & Public)
- Bit Bucket Projects
- Tokens

- Push
- Pull
- Cloning
- Branch
- Merge

BUILD TOOLS - MAVEN

- Introduction
- Maven Goals
- What Maven Does
- What is Build Tool?
- pom.xml
- Requirements for Build
- Maven Architecture
- Maven Build Life Cycle
- Maven Directory Structure
- Generating Jar File
- Problems Without Maven
- Maven vs Ant
- Interview Questions

INTEGRATION TOOL - JENKINS

- Introduction
- Workflow
- Continuous Integration
- Continuous Delivery
- Continuous Deployment
- Jenkins Setup
- Jenkins Setup Requirements
- Java Installation
- Git Integration
- Maven Integration
- Jobs Types in Jenkins
- Free Style Job
- Pipeline



- Maven Job, Task
- Parameter Building
- Choice Parameter
- File Parameter
- String Parameter
- Multi-String Parameter
- Cron Jobs
- Poll SCM
- Webhooks
- Linked Projects
- Up Stream
- Down Stream
- Build Pipeline View
- Master – Slave Architecture
- Master – Slave Setup
- Environment Variables
- Local Variable
- Global Variable
- Disabling Jobs
- Renaming Jobs
- Build Pipeline View
- Shell Execute
- Build Triggers
- Authentication for Build Trigger
- Pipelines
- Scripted Pipelines
- Declarative Pipelines
- Pipeline Stages
- Pipeline Syntax
- Staged Pipeline
- Multi-Staged Pipeline
- Pipeline as a Code
- Multiple Commands on Pipeline
- PAAC Variables
- PAAC Input Parameter
- PAAC Postbuild
- PAAC Input
- Pipeline Through Build Pipeline

- Pipeline Post-Build
- Plugin Management
- Console Output
- List View
- Custom View
- Build History
- Cron Syntax Generator
- Editing Build Info
- Agent Setup
- User Management
- Adding Users
- Assigning Roles
- Managing Roles
- Multi Branch Pipelines
- Interview Question

CONTAINERIZATION TOOL - DOCKER

- Microservices Introduction
- Difference Between Monolithic & Microservices
- Micro Services Advantages
- Micro Services Limitations
- Virtualization Intro
- Docker vs VMs vs Servers
- Before Docker vs After Docker
- Docker Introduction
- Understanding Docker
- Docker Architecture
- Docker Client
- Docker Server
- Docker Daemon
- Docker Objects
- Docker Basic Commands



- Stop vs Kill
- Run vs Cmd vs Entrypoint
- Add vs Copy
- Run vs Pull
- Docker Exec
- Container Limits
- Docker Images
- Docker Containers
- Docker Volumes
- Working With Containers
- Build Image From Container
- Creating Custom Images
- Ways of Container Creation
- Docker File
- Docker File Components
- Docker File Creation
- Docker Volumes
- Uses of Volumes
- Creating Volumes
- Creating Volumes By Using Commands
- Volumes (Container - Container)
- Volumes (Host - Container)
- Creating Volume From File
- Creating Volume Form Commands
- Volume Mounting
- Volume Modifying
- Docker Port Mapping
- Docker Multistage Build
- Creating Jenkins By Docker
- Docker Registry Types
- Cloud Based Registry vs Local Registry
- Docker Login
- Docker Hub Account Creation
- Image Pushing To Registry
- Docker Swarm
- Docker Swarm Architecture
- Docker Swarm Components
- Docker Swarm Manager

- Docker Swarm Worker
- Docker Swarm Service
- Docker Portainer
- Docker Compose
- Docker Compose Advantages
- Docker Compose Installation
- Docker Compose File
- Docker Compose File in JSON
- Docker Compose File in YAML
- Docker Compose Single Service
- Docker Compose Multi Services
- Docker Compose File Commands
- Docker Compose Scale
- Docker Stack
- Portainer Introduction
- Portainer Use Cases
- Portainer Setup
- Working With Portainer
- Memory Managemnet
- ECS Taks & Services
- Load Balancers
- Hosting an Application
- Docker Networks
- Why to Use Docker Networks
- Types of Networks
- Interview Questions

KUBERNETES

- History
- Container Orchestration
- Kubernetes Introduction
- Architecture
- Master Components
- Node Components
- Working With K8s



- Role of Master
- Components of Control Plane
- Kube-API Server
- ETDC
- Features
- Kube-Scheduler
- Control Manager
- Node Components
- Kubelet
- Container Engine
- Kube-Proxy
- Pod
- Pod Limitations
- Pod Multi Container Pod
- Minikube Intro
- Single Node Cluster
- ReplicaSet
- Drawback of Replica Set
- DaemonSet
- Drawback of Daemon Set
- Deployments
- Labels & Selectors
- Rolling Updates
- Scaling Pods
- Types of Scaling
- Deployment Scaling
- Pausing & Unpausing Rolling Deployment
- Proportional Scaling
- HPA
- Kops
- Kops Adatnages
- Kops Installation
- Kubectl Installation
- Cluster Creation Using Kops
- Cluster Scaling
- Service in Kubernetes
- Types of Services
- Cluster IP

- Node Port
- Load Balancer
- Ingress
- NodePort vs ClusterIP vs LB
- Kubernetes Volumes
- Empty Dir
- Hostport
- Persistent Volume
- Persistent Volume Claim
- What is Probes
- Use Cases of Probes
- Readiness Probe
- Liveness Probe
- Startup Probe
- Readiness vs Liveness vs Startup
- Config Maps
- Secrets
- What is RBAC
- Namespaces
- Resource Quotas
- Helm
- Statefull Application
- Stateless Application
- Statefull va Statless Application
- Working With Stateful Applications
- Working With Stateless Applications
- Prometheus & Grafana
- Interview Questions

MONITORING : DATADOG

- What is Datadog
- How to Setup Datadog
- How to Monitor App Servers
- Monitor Jenkins Pipelines
- Monitor Containers
- Export Logs to Datadog



MONITORING : NEW RELIC

- What is New Relic
- How to Setup New Relic
- How to Monitor Containers
- Setting Up Mail Notifications and Events
- Export Logs to New Relic

CONFIGURATION

MANAGEMENT TOOL - ANSIBLE

- Ansible Introduction
- Ansible Usecases
- Ansible History
- Ansible Architecture
- Ansible Inventory Host Pattern
- Ansible Execution Types
- Ad-Hoc Commands
- Working With Adhoc
- Ansible Modules Introduction
- Working With Ansible Modules
- Ansible Playbooks
- YAML Introduction
- YAML Indentation
- Commenting Sections
- Ansible Variables
- Ansible Handlers
- Ansible Loops
- Ansible Conditions
- Ansible Vault
- Ansible Roles
- Ansible Tags
- Ansible Dry Run
- Ansible Pip
- Ansible Userinfo
- Ansible Setup Modules

- Ansible Debug Module
- Ansible Raw Module
- Ansible Command Module
- Ansible Copy Module
- Ansible Shell
- Static Website Using Playbook
- Advantages & Disadvantages
- Interview Questions

INFRA AS A CODE TOOL - TERRAFORM

- Introduction
- History
- Advantages & Dis Advantages
- Terraform Setup & Installation c
- Terraform Init
- Terraform Plan
- Terraform Apply
- Terraform Destroy
- Creating a main.tf
- Role Based Authentication
- S3 Backend Setup
- Terraform Variables
- String
- Number
- Boolean
- List
- Terraform Loops
- Terraform Workspaces
- Terraform Locals
- Terraform Outputs
- Creating VPC
- Creating EC2
- Creating S3 Bucket
- Creating Security Groups



- Creating Subnets
- Creating EBS
- Terraform Multiple tfvar Files
- Terraform CLI
- Dynamic Block

CODE & ARTIFACT STORAGE

- Nexus
- SonqrQube

INTEGRATIONS

- Git Integrated with Jenkins
- Maven Integrated with Jenkins
- Ansible Integrated with Jenkins
- Docker Integrated with Jenkins
- Terraform Integrated with Jenkins
- S3 Integrated with Jenkins
- Nexus Integrated with Jenkins
- Sonar Integration with Jenkins
- Slack Integration with Jenkins
- Splunk Integration with Jenkins

SECURITY TOOLS

- Sonarqube
- Open Source Web Application Secure Project
- Trivy
- Kube bench
- Istio
- Calico
- AWS Backup
- Terraform Vault
- TF Sec

AI WITH DEVOPS :

- Integrating GitHub Copilot for generating files such as Manifest, Jenkinsfile, Dockerfile, and Terraform files
- Introduction to Harness as a CI tool
- Using Cast AI for monitoring and cost optimization of Kubernetes clusters



TOP CITIES FOR DEVOPS WITH MULTI-CLOUD (AWS+AZURE)

- BANGALORE (BENGALURU)
- GURGAON (GURUGRAM)
- HYDERABAD
- PUNE
- MUMBAI
- CHENNAI
- NOIDA

KEY SECTORS DRIVING DEMAND FOR DEVOPS WITH MULTI-CLOUD (AWS+AZURE)

- TECHNOLOGY AND SOFTWARE DEVELOPMENT
- E-COMMERCE HEALTHCARE
- FINANCE AND FINTECH
- HEALTHCARE
- EDUCATION AND EDTECH
- DATA SCIENCE AND ANALYTICS
- ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
- MEDIA AND ENTERTAINMENT
- STARTUPS AND TECH INCUBATORS



TOP COMPANIES HIRING

Infosys

tcs TATA
CONSULTANCY
SERVICES

accenture


cognizant

Capgemini

wipro

IBM

BOX8

KPMG



ALL THE BEST





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